

Requirements Document — Hospital Inventory Management System

Project Overview

City General Hospital requires a centralized inventory management system to track medical supplies, pharmaceuticals, and equipment across 12 departments. Current process is spreadsheet-based, causing stockouts and waste due to expiry.

The system must integrate with the existing Oracle ERP and support barcode/RFID scanning via mobile devices.

Actors

Inventory Manager: manages stock levels, purchase orders, and supplier relationships.

Department Nurse: requests supplies, reports usage, flags shortages.

Pharmacy Staff: manages pharmaceutical inventory with lot tracking and expiry dates.

Finance Officer: approves purchase orders, reviews cost reports.

System Admin: manages user access, integrations, and audit logs.

Functional Requirements

FR-1 [High] The system shall track real-time stock levels per item, per department, with automatic low-stock alerts.

FR-2 [High] Staff shall scan barcodes or RFID tags via the mobile app to log consumption.

FR-3 [High] The system shall auto-generate purchase orders when stock falls below the reorder threshold.

FR-4 [High] Pharmaceuticals shall be tracked by lot number and expiry date with 90-day expiry alerts.

FR-5 [Medium] The system shall sync purchase orders bidirectionally with Oracle ERP via REST API.

FR-6 [Medium] Finance officers shall approve or reject purchase orders above \$10,000 via in-app workflow.

FR-7 [Low] The system shall generate monthly waste and cost-savings reports in PDF and Excel.

Non-Functional Requirements

NFR-1 [High] Mobile app must work offline and sync when connectivity is restored.

NFR-2 [High] Data retention for 7 years to comply with hospital accreditation requirements.

NFR-3 [Medium] System must support 200 concurrent mobile users without performance degradation.

Technical Constraints

Must deploy on-premise within the hospital data center (no cloud PHI storage).

Mobile app must support Android 10+ and iOS 15+ on hospital-issued devices.

Oracle ERP version 12.2 — API integration only, no direct DB access permitted.